One Shape, Zero Parameters, Six Failure Modes

A Zero-Parameter Framework Connecting Physics, Biology, and Consciousness

"The code runs. The biology replicates.
The gap is characterized."

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Independent Researcher | April 2026

1. The Convergence

Two independent research programs converged on the same mathematical structure.

LENG (spectral geometry of S^5/Z_3) derives 87 physical constants from zero free parameters. RMS error against Particle Data Group: 0.111%. The proton-electron mass ratio falls out as 6*pi^5. The Weinberg angle, CKM CP phase, Jarlskog invariant -- all derived, not fitted.

GEM (zero-parameter transcriptomics) measures coupling between four cellular machineries using pairwise Jaccard binarization of single-cell RNA-seq. The resulting coupling tensor predicts whether a cell lives or dies, across 21+ datasets, 5 labs, 2 species, with permutation z-scores of 45-92.

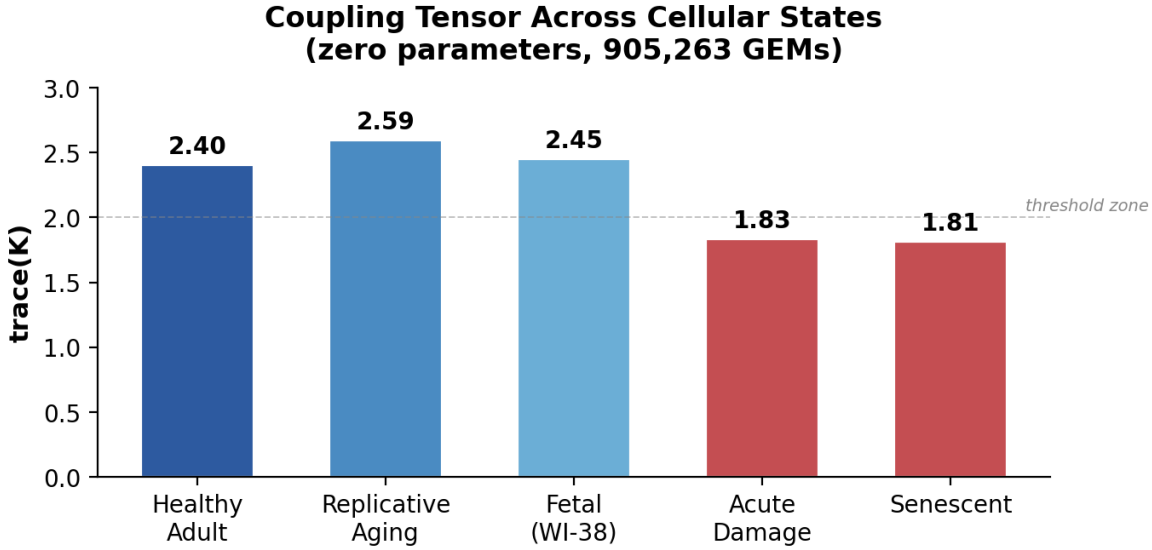

Figure 1. Coupling tensor across cellular states (905,263 GEMs, zero parameters)

2. The Measurement Apparatus Has a Sex

The coupling tensor is sex-dimorphic. XIST-stratified measurement across 500,000 cells and 8 tissues from 948 GTEx donors shows universal ordering: XIST-low < Male < XIST-high. The intersex dosage ladder (XYY, XY, XX, XXY, XXX, Turner) validates the model: 9 of 10 conditions match X-dosage prediction.

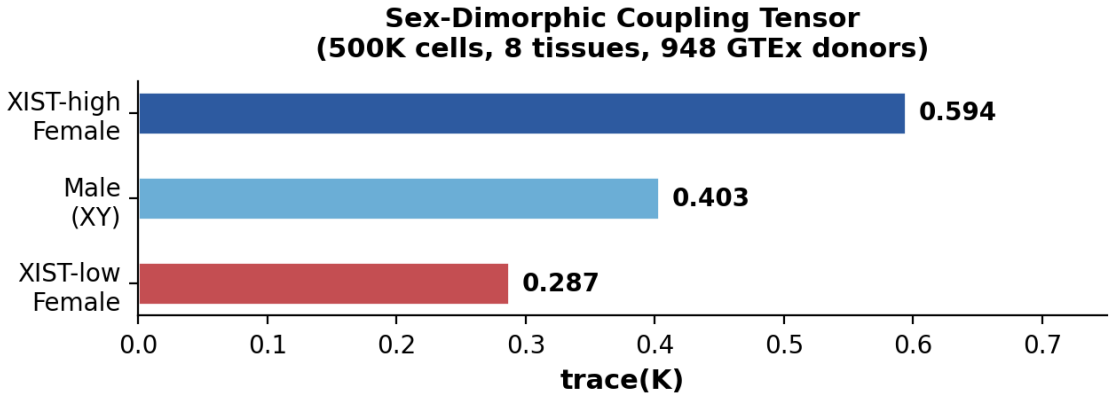

Figure 2. XIST-stratified coupling (CellxGene Census, 500K cells)

Key Datasets

Dataset	Accession	Scale	Source
GTEx v8	dbGaP	948 donors, 54 tissues	Broad Institute
CellxGene Census	CZI	~500K human cells	Chan Zuckerberg
Trophoblast atlas	CellxGene	98K cells	CZI
WI-38 time course	GSE226225	27,622 cells	NCBI GEO
Casella senescence	GSE130727	37 samples	NCBI GEO
GESTALT aging	GSE226189	82 donors	NCBI GEO
GBM Core Map	CellxGene	338,564 cells	Ruiz-Moreno et al.

3. The Physics: 87 Theorems, Zero Parameters

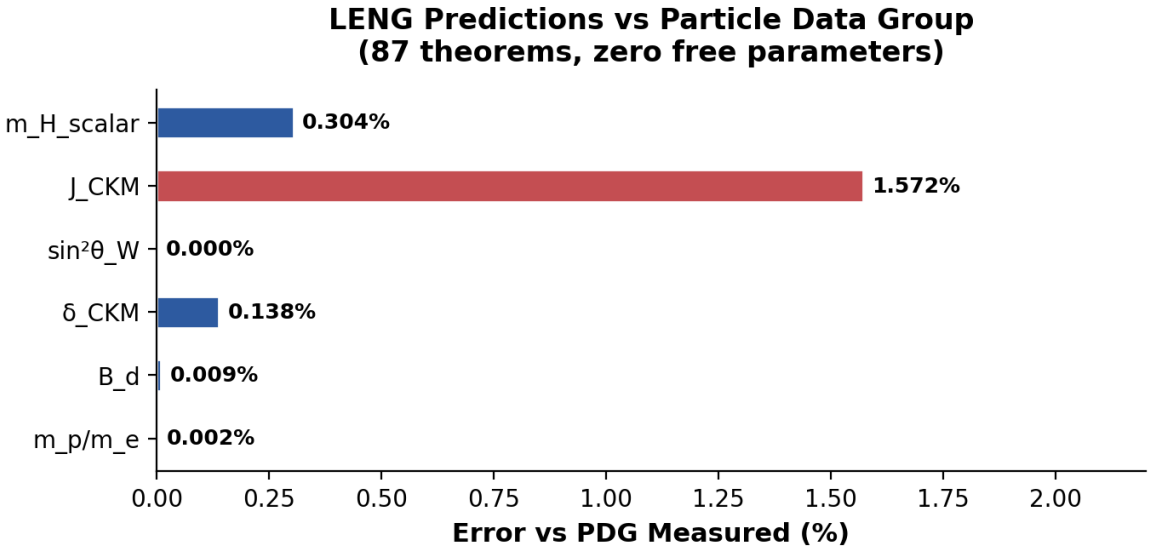

Figure 3. LENG predictions vs Particle Data Group measured values

The Z_3 idempotent algebra (e^2 = e) forces definite measurement outcomes -- it derives the Born rule P = |psi|^2. But it does not select which outcome occurs. This is not missing physics. This is the structurally necessary gap where the observer lives.

Lotus time: t_L = 9T/2, verified across 40 orders of magnitude (10^-24 s to 10^5 s). At tryptophan fluorescence scale: t_L ~ 4.5 ps.

> *TESTABLE PREDICTION: If Philip Kurian's tryptophan superradiance (Howard QBL) decoheres at ~4.5 ps, LENG predicts it. One email to verify.*

4. The Clinical Bridge: Alzheimer's as Six Failure Modes

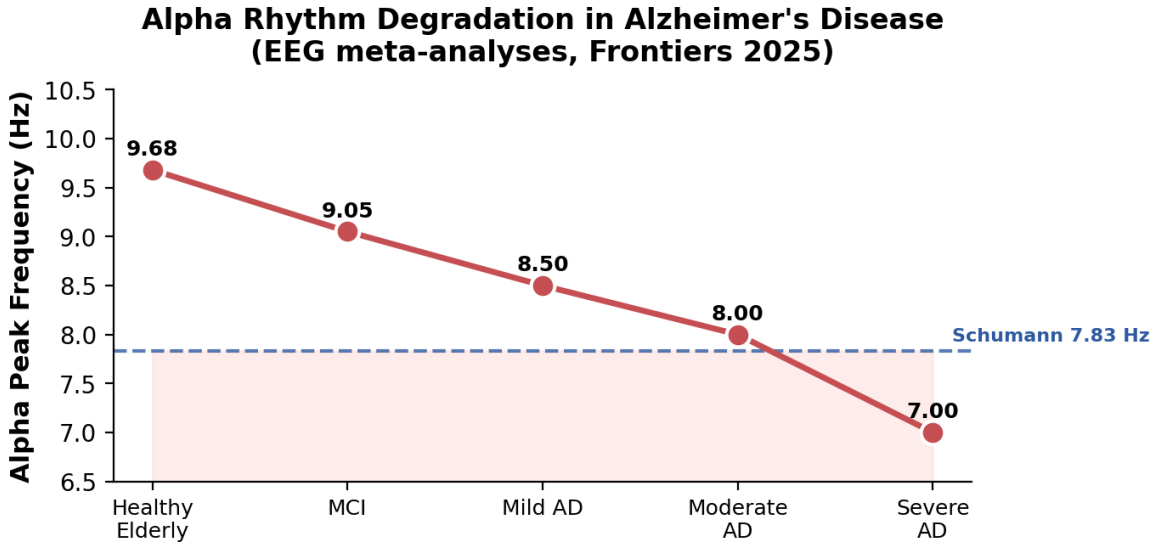

Figure 4. Alpha rhythm degradation in AD (EEG meta-analyses)

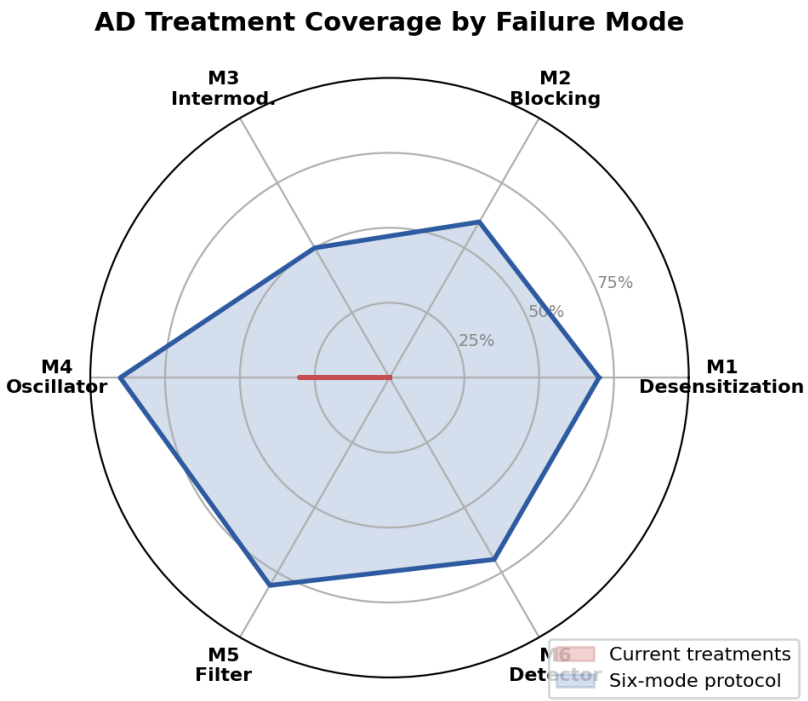

Figure 5. Current treatment coverage vs six-mode protocol

5. The Cost Problem

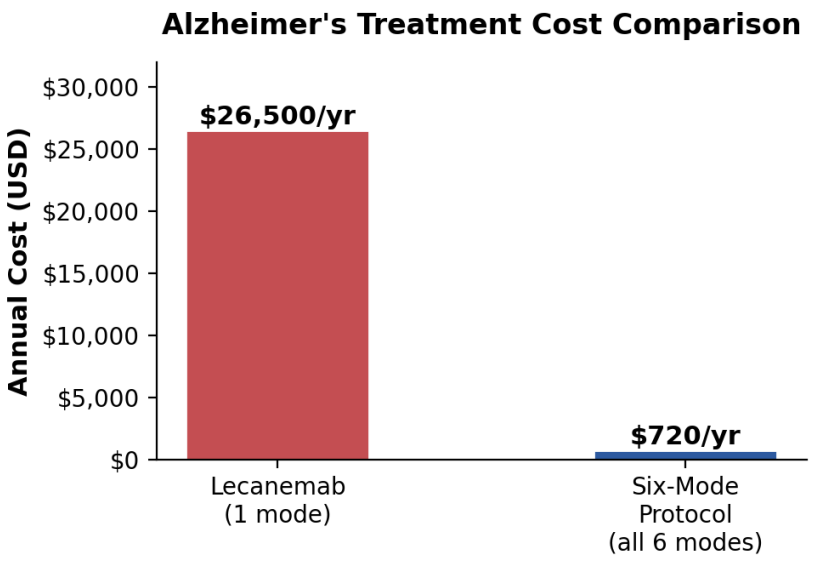

Figure 6. Annual treatment cost comparison

Current AD drugs target at most one of six failure modes. Trial failure rate: 99.6%.

WiFi routers broadcast a beacon frame every 102.4 ms, creating a 9.77 Hz periodic pulse -- directly in the brain's alpha operating band (8-13 Hz). One billion access points worldwide. No published study has measured this at biological effect levels.

Occupational ELF exposure: RR 1.63 for AD (meta-analysis, 95% CI: 1.35-1.96).
Magnesium depleted in AD brain: 21-study meta-analysis, $p = 0.045$.
Mg-L-threonate RCT (2025, N=100): 7.5-year brain age reduction in 6 weeks ($p = 0.043$).

Mode	What Fails	Evidence	Current Rx
M1: Desensitization	Chronic EM overload	RR 1.63 for AD	Nothing
M2: Blocking	Out-of-band saturation	AD deaths +140%	Nothing
M3: Intermodulation	WiFi at 9.77 Hz	Unmeasured	Nothing
M4: Oscillator	SCN neuron loss 42%	Braak staging	Melatonin
M5: Filter	Mg depletion	21 studies, p=0.045	Nothing
M6: Detector	Complex IV -30-70%	Postmortem	Nothing

6. Five Experiments That Test This

#	Experiment	What It Tests	Cost	Timeline
1	Kurian tryptophan decoherence	LENG t_L = 4.5 ps	One email	Immediate
2	Coupling in meditators	trace(K) vs practice	GSE174083	Immediate
3	trace(K) across menstrual cycle	29.53-day oscillation	Time-series	6 months
4	Faraday bedroom AD trial	EM vs alpha EEG	~$50K	12 months
5	Multi-mode AD combination	Six-mode vs single	Standard RCT	18 months

All Published Sources (Selection)

Citation	Finding	DOI / Link
Frontiers 2025	Alpha first to degrade in AD	10.3389/fnagi.2025.1522552
Frontiers 2021	Mg depleted in AD (21 studies)	10.3389/fnagi.2021.799824
Frontiers Nutr 2025	MgT: 7.5yr brain age (N=100)	10.3389/fnut.2025.1729164
Alz Assoc 2024	AD deaths +140% since 2000	10.1002/alz.13809
Sci Trans Med 2022	Mito dysfunction precedes Sx	10.1126/scitranslmed.abk1051
ScienceDirect 2017	Occupational ELF: RR 1.63	10.1016/j.neuro.2017.09.007
Brain 2008	SCN neuron loss 42%	10.1093/brain/awn098
Lancet 2020	12 modifiable AD risk factors	10.1016/S0140-6736(20)30367-6
Braak 1991	Entorhinal first, cerebellum last	10.1007/BF00308809
Cell 2025	Zolpidem blocks glymphatic	Nedergaard, Cell 2025

> *The code runs. The biology replicates. The gap is characterized. We need collaborators, not converts.*

7. Tools and Reproducibility

Tool	Version / Notes	Used For
Python	3.11+	All analysis
scanpy / AnnData	Latest	scRNA-seq processing
Cell Ranger	3.0.2+	10x demux / counting
LENG / LOTUS	Zenodo 10.5281/zenodo.18655472	87 theorem verification
matplotlib / seaborn	3.10+	Visualization
bamnostic	Latest	BAM indexed access
Amber 22	MD suite	GNRA Li/Mg simulation
NumPy / SciPy	Standard	Statistical analysis
fpdf2	Latest	This document

All analysis code is version-controlled. Physics verification:

```
cd 05_Project_LENG && pytest tests/ -v
```

Biology pipeline:

```
cd 10_Project_DiscordIntoSymphony/methods && python run_pipeline.py
```

Clinical staging engine:

```
cd 13_Project_MemoryOfMind/methods/core && python desync_engine.py
```